Rohan Batra

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SUMMARY

Backend engineer building containerized Python services — FastAPI, SQLAlchemy, Redis, and MongoDB — deployed on AWS with CI/CD and infrastructure as code.

SKILLS

Languages: Python, TypeScript, JavaScript, Shell, C, C++, Swift, TeX
Backend: Flask, FastAPI, Pydantic, Uvicorn, Alembic, SQLAlchemy, Android SDK, Hugo, Microservices, REST APIs
Data: MongoDB, Redis, PostgreSQL, ChromaDB, MySQL, DuckDB, DuckLake
Infrastructure: Docker, Helm, Nginx, Proxmox, Terraform, ADB, Evolution X, Fastboot, Garuda Linux, Kali NetHunter
DevOps: GitHub Actions, MkDocs, Obsidian, Termux, cargo-deny, cargo-llvm-cov, cargo-mutants
AI/LLM: Claude, Anthropic API, LangChain, Ollama, Agent Evals & Guardrails, Agent Hooks, Agent Skills, Anthropic Messages API
Security: Gitleaks, HashiCorp Vault, Semgrep, Trivy, age encryption, AppArmor, Burp Suite, Cerbos
Networking: DHCP / NAT, DNS, IP Allocation & Subnetting, pfSense, Pi-hole, NetBird
Frontend: CSS, Next.js, React, Tailwind CSS, HTML, Dart

PROJECTS

Paridyum — Independent Multi-Tenant Platform *Rust, OpenTofu, Ansible, Talos Linux, Kubernetes, Cilium, Next.js, PostgreSQL*
Self-directed systems project: built solo from an empty directory to a running five-node Kubernetes cluster on self-hosted metal with a public internet edge.

- Built a **specification-as-software governance system**: 251 versioned artifacts (constitution, architecture decision records, standards, RFCs) with machine checks mapped outward to **NIST SSDF, CIS v8.1 and NIST CSF 2.0**, a multi-axis test-type taxonomy, and a precedence order in which existing code is the lowest authority.
- Delivered the full infrastructure path: hardened bare metal (AppArmor, nftables, confined QEMU/libvirt) through **Proxmox and OpenTofu** to **Talos Linux and Kubernetes** with Cilium as CNI and kube-proxy replacement, L2-announced load-balancer VIPs, and etcd snapshots that assert they captured something.
- Wrote **31,000+ lines of Rust** across two production tools — hexagonal architecture, trait-based abstraction, injectable clock adapters, and three independent SHA-256 hash chains for tamper-evident audit records.
- Engineered a **governable AI development harness**: a Claude Code plugin whose PreToolUse hooks make violating a locked architectural decision impossible rather than discouraged, with class-filtered context injection and append-only research provenance.

UPES-ECS — Campus Emergency Communication System *Juniper SRX/EX/Mist, Asterisk/PJSIP, FastAPI, QEMU, PowerShell*
LAN-only emergency calling on HPE Juniper networking. Presented to Team HPE at their Bengaluru campus and at UPES, July 2026.

- Built a **LAN-only** campus emergency calling system (one-number 111 hotline) on Asterisk/PJSIP requiring no internet, SIM, or cloud — designed, deployed and operated end-to-end on an **HPE Juniper** networking backbone.
- Configured the enterprise fabric by name: **Juniper SRX300/320** as the guarded edge (firewall, DHCP, NTP, LAN-only boundary), a **Virtual Chassis of two EX2300-C-12P** PoE switches carrying the voice network, and two **Mist AP32** Wi-Fi 6 access points for campus-wide softphone reach.
- Engineered the voice fabric for zero-touch operation and self-healing: **LLDP-MED** auto-provisioning of voice VLAN, PoE class and DSCP; strict-priority **CoS for RTP**; **DHCP-snooping with Dynamic ARP Inspection** to protect the failover path; and Junos **event-options** policies that power-cycle a hung AP without human intervention. Budgeted PoE against the switch's real 124 W ceiling.
- Designed **offline caller location by switch port** (IP to MAC to port to room) so the network identifies where a 111 call originated with no cloud or location service, and fed live switch/AP state to the Operations Console over **NETCONF**.
- Operated it as a production system where downtime is a safety defect — active/standby high availability, health checks, load testing, and incident root-cause analysis. Three documented RCAs: a serialized single-threaded proxy, a NAT media-address fault dropping every call at 32 seconds, and leaked channels with no teardown path.
- Developed a FastAPI status/control API exposing real-time queue, call, and system state to an operations console and live LED-TV wallboards, provisioned end-to-end via a single PowerShell installer.
- Implemented multilingual IVR across **44 languages** with Piper TTS voice prompts and an offline panic-coach fallback, gated by CI (Ruff, ShellCheck, PSScriptAnalyzer, gitleaks). Presented to **Team HPE** at HPE's Bengaluru campus (15 Jul 2026) and demonstrated live at UPES (20 Jul 2026).

Anvil — Agentic Revenue-Recovery Control Plane *Python, FastAPI, LangGraph, PostgreSQL, Claude, Alembic*
Recovery of failed recurring payments where the model decides and the ledger disposes.

- Designed an agentic revenue-recovery system for failed recurring payments where an LLM proposes recovery actions

and a deterministic policy engine, mandate registry, and append-only ledger dispose — no model call can move money directly. Built for the **Razorpay AI Buildathon 2026** (Track 03).
- Implemented a dynamic-programming retry scheduler over calibrated hazard curves to pick the optimal retry time for a failed debit, alongside a **27-rule policy engine** with LangGraph-orchestrated pause/resume for human approval and step-up authentication.
- Enforced **10 financial invariants** — append-only ledger, zero-sum postings, idempotent webhooks, row-locked budget reservations — at the database level, backed by 203 tests.

Second Brain Database *Python, FastAPI, MongoDB, Qdrant, Redis, Celery, Docker, Helm*
Self-hosted personal knowledge management platform, containerized and multi-tenant.

- Architected a **474,000-line** multi-service Python backend on FastAPI with MongoDB, Qdrant vector search, Redis, and Celery for asynchronous work.
- Built secure REST APIs with **OAuth2 authentication and role-based access control (RBAC)**, with optimized schemas and Redis caching.
- Containerized the full stack with Docker and Helm, with separate dev/test/prod GitHub Actions pipelines gating every build, and built a supporting ecosystem of **12 Next.js microfrontends** plus a Flutter mobile client.

EXPERIENCE

Hardware & WireGuard Research Intern *2026-07 – Present*
Exaflair Technologies Pvt. Ltd.

Platform Engineering Intern *2025-11 – 2026-07*
Eve

- Founded and led the **Eve Data Management Service**: a FastAPI microservice with source/destination/connection APIs, SQLModel models, Alembic migrations, an Airbyte Config API client, sync orchestration, webhook routes, health checks, and Kubernetes/EKS deployment.
- Implemented **schema-per-tenant multi-tenancy** using PostgreSQL search_path switching, with tenant context propagation and tenant-aware service and database isolation.
- Built **Airbyte-based ingestion** — typed connector definitions, USER vs PLATFORM destination modes, connection testing, sync polling and history tracking, and S3/MinIO bucket auto-creation — and resolved staging failures across Airbyte URLs, timeouts, secrets, readiness probes, and Temporal support.

Technical Head *2025-09 – Present*
Google Developer Groups On Campus (GDGOC), UPES Dehradun

- Led and delivered technical workshops for **100+ participants** across cloud, system design, and full-stack development.
- Managed cross-functional teams executing production-grade initiatives.

Linux Support & Community Volunteer *2023-09 – 2025-04*
LinuxWale, Dehradun

- Onboarded **10+ students** to Linux — distribution selection, dual-boot setup, and driver/hardware troubleshooting across varied hardware.
- Wrote shell scripts to automate post-install configuration and routine setup tasks.

EDUCATION

UPES Dehradun — B.Tech (Hons.) Computer Science Engineering, Cyber Security *2023-08 – 2027-08 · CGPA 8.1/10*

ACHIEVEMENTS

- **Top 15% — TryHackMe cybersecurity learning platform**
- **Certificate of Appreciation — Golden Key Eagles Women Empowerment Center (Sep 2024)**
- **Certificate of Appreciation — HQ Uttarakhand Sub Area (Aug 2024)**
- **Certificate of Appreciation — Golden Key Tuskers, AWWA Week 2024 (Aug 2024)**
- **Semifinalist — Eureka! Junior 2022–23, E-Cell IIT Bombay (Issued Nov 2022; Credential ID: ecell2021)**
- **Semifinalist — Eureka! Junior 2021–22, E-Cell IIT Bombay (Issued Dec 2021; Credential ID: ecell2021)**